

```

15.
16. View this code at https://nostarch.com/big-book-small-python-projects
17. Tags: large, artistic"""
18.
19. import shutil, sys
20.
21. # Set up the constants for line characters:
22. UP_DOWN_CHAR = chr(9474) # Character 9474 is '|'
23. LEFT_RIGHT_CHAR = chr(9472) # Character 9472 is '─'
24. DOWN_RIGHT_CHAR = chr(9484) # Character 9484 is '┐'
25. DOWN_LEFT_CHAR = chr(9488) # Character 9488 is '└'
26. UP_RIGHT_CHAR = chr(9492) # Character 9492 is '┌'
27. UP_LEFT_CHAR = chr(9496) # Character 9496 is '┐'
28. UP_DOWN_RIGHT_CHAR = chr(9500) # Character 9500 is '└─'
29. UP_DOWN_LEFT_CHAR = chr(9508) # Character 9508 is '─┐'
30. DOWN_LEFT_RIGHT_CHAR = chr(9516) # Character 9516 is '└─┐'
31. UP_LEFT_RIGHT_CHAR = chr(9524) # Character 9524 is '┌─┐'
32. CROSS_CHAR = chr(9532) # Character 9532 is '┘'
33. # A list of chr() codes is at https://inventwithpython.com/chr
34.
35. # Get the size of the terminal window:
36. CANVAS_WIDTH, CANVAS_HEIGHT = shutil.get_terminal_size()
37. # We can't print to the last column on Windows without it adding a
38. # newline automatically, so reduce the width by one:
39. CANVAS_WIDTH -= 1
40. # Leave room at the bottom few rows for the command info lines.
41. CANVAS_HEIGHT -= 5
42.
43. """The keys for canvas will be (x, y) integer tuples for the coordinate,
44. and the value is a set of letters W, A, S, D that tell what kind of line
45. should be drawn."""
46. canvas = {}
47. cursorX = 0
48. cursorY = 0
49.
50.
51. def getCanvasString(canvasData, cx, cy):
52.     """Returns a multiline string of the line drawn in canvasData."""
53.     canvasStr = ''
54.
55.     """canvasData is a dictionary with (x, y) tuple keys and values that
56.     are sets of 'W', 'A', 'S', and/or 'D' strings to show which
57.     directions the lines are drawn at each xy point."""
58.     for rowNum in range(CANVAS_HEIGHT):
59.         for columnNum in range(CANVAS_WIDTH):
60.             if columnNum == cx and rowNum == cy:
61.                 canvasStr += '#'
62.                 continue
63.
64.             # Add the line character for this point to canvasStr.
65.             cell = canvasData.get((columnNum, rowNum))
66.             if cell in (set(['W', 'S']), set(['W']), set(['S'])):
67.                 canvasStr += UP_DOWN_CHAR
68.             elif cell in (set(['A', 'D']), set(['A']), set(['D'])):
69.                 canvasStr += LEFT_RIGHT_CHAR

```